Mobility of Morocco[1]

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The mobility of Morocco relative to Africa during the early history of the Atlantic has long been debated: arguments have been developed from Atlantic kinematic considerations and from paleomagnetic results. Both types of arguments are reexamined here. Using a new model of the Atlantic Ocean evolution described elsewhere, it is shown that the reconstructions of the positions of Africa relative to North America before the Atlantic opening, and at the times of magnetic anomalies M22 and J, do not imply a major motion of Morocco independent of Africa during these periods of time. The corresponding geomagnetic paleopoles have been recomputed from sample sites located on both "mobile" Morocco and "stable" Africa. The results indicate that the virtual geomagnetic pole of "mobile" Morocco for the Liassic falls within the 95% confidence cone of "stable" Africa. It is thus concluded that no major movement has occurred between "mobile" Morocco and "stable" Africa during the early phases of opening of the central Atlantic Ocean. This is in accordance with the field geological observations on the South Atlas fault; however, limited motion along this lineament, as observed in the field, is still compatible with the above conclusion, owing to the limited resolving power of both kinematic and paleomagnetic methods.

La mobilité du Maroc par rapport à l'Afrique au début de l'histoire de l'Atlantique, a déjà donné lieu à de longs débats: les arguments utilisés viennent soit des contraintes imposées par la cinématique de l'Atlantique, soit des résultats du paléomagnétisme. Ces deux types d'arguments sont passés en revue dans ce travail. En se fondant sur un modèle nouveau, décrit ailleurs, de l'évolution de l'océan Atlantique, on montre que la restitution des positions relatives de l'Afrique et de l'Amérique du Nord avant l'ouverture de l'Atlantique et à l'époque des anomalies magnétiques M22 et J, n'implique pas un mouvement majeur du Maroc indépendamment de l'Afrique aux périodes considérées. Les paléopoles géomagnétiques correspondants ont été apparaît que le pôle géomagnétique virtuel du Maroc "mobile" tombe à l'intérieur du cône de confiance à 95% du pôle de l'Afrique "stable." On en conclut donc qu'aucun mouvement majeur entre Afrique "stable" et Maroc "mobile" n'a eu lieu durant les premières phases de l'ouverture de l'Atlantique central. Ceci est en accord avec les observations géologiques de terrain sur la faille sud-atlasique; cependant des mouvements limités le long de ce linéament, pour lesquels il existe des indices sur le terrain, ne sont pas incompatibles avec la conclusion ci-dessus, puisque la précision des méthodes cinématiques et paléomagnétiques utilisées reste très limitée.

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Introduction

South from its northernmost Alpine region (Rif), Morocco is comprised of (Fig. 1) (see, e.g., Michard 1976) the Moroccan meseta (plus the western part of the Oranese meseta), affected by Hercynian tectonism, on which a rather thin and tabular Mesozoic cover overlies a Paleozoic basement which outcrops over large areas, and the Precambrian–Paleozoic Saharan platform, the northern border of which (Anti-Atlas) was deformed in Hercynian times.

The High- and Middle-Atlas domains experienced extension at the beginning of the Mesozoic Era: two subsident basins formed and 3000–4000 m of Jurassic and Cretaceous sediments were deposited. The most active subsidence took place during the Jurassic Period. At the Jurassic–Cretaceous transition, the domain was affected by an important compressive episode; at the end of the Jurassic it

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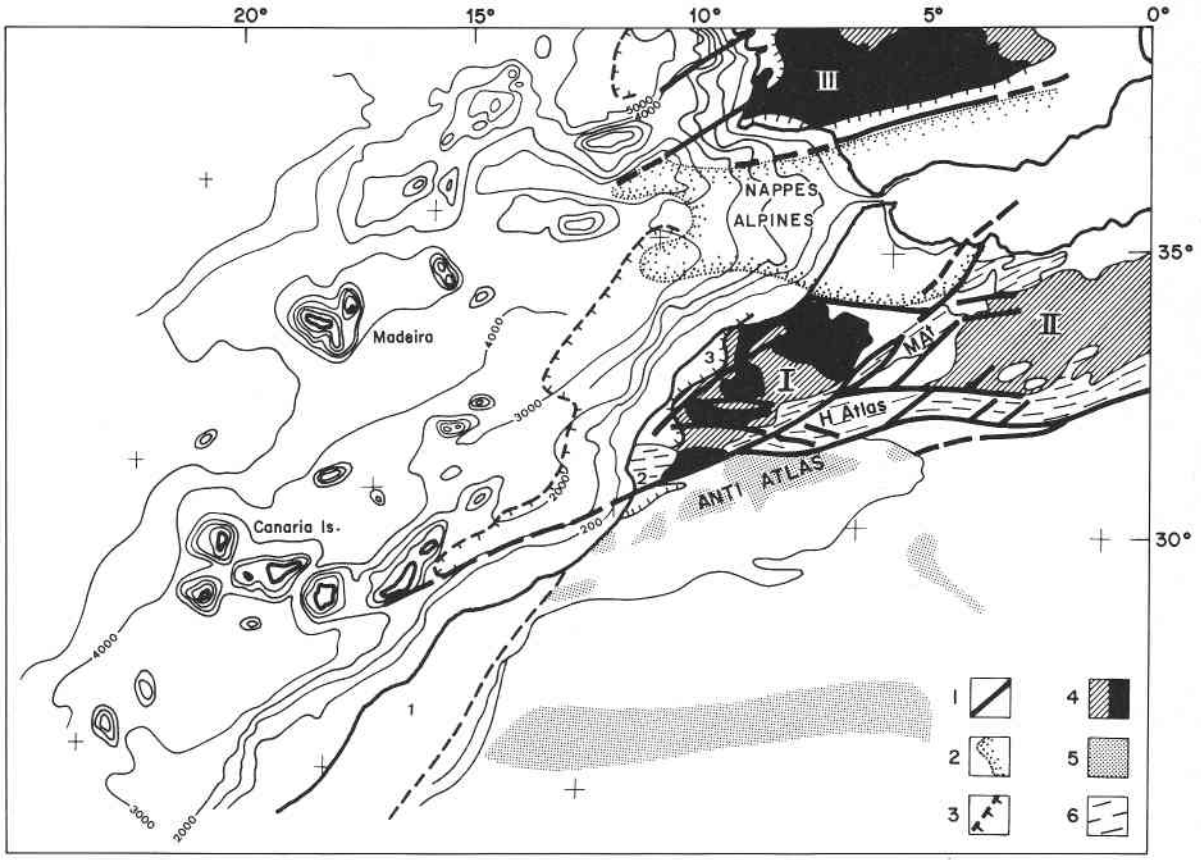


FIG. 1. Geological structural sketch map. (1) Tardi-Hercynian wrench-fault pattern after Mattauer *et al.* (1977). (2) Betico-rifan domain. (3) Limits of the Mesozoic basins: (1) Rio de Oro; (2) Essaouira; (3) West Moroccan. At sea, extension of Triassic–Liassic evaporitic basins, after Grunau *et al.* (1975) and Bonnin (1978). (4) Tabular Paleozoic and Mesozoic formations: (I) Moroccan meseta; (II) Oranese meseta; (III) Iberian meseta. (5) Precambrian. (6) Atlas chains.

emerged, owing to compression, and the compressive conditions persisted up to the Quaternary (Mattauer *et al.* 1977).

The limit between the Anti-Atlas and the High-Atlas (and farther to the east) is marked by a major fracture, the South Atlas fault (Rod 1962), which de Sitter (1964) regarded as one of his "great fundamental faults." To the west, in the Atlantic Ocean, the South Atlas fault is in continuity with the Canary Island volcanic lineament, through a marked change in direction of the continental shelf break.

Many authors have suggested that this fault played a major role in the assumed motion of several tens of kilometres of the meseta relative to the west African block, on either paleomagnetic or Atlantic kinematic grounds. We examine below, succesively and critically, both kinds of arguments and conclude that there was nonexistent or very limited differential motion after anomaly M22 (approximately 150 Ma).

Atlantic Kinematics

Pitman and Talwani (1972) have worked out a reconstruction of the central Atlantic at approximately 150 Ma, i.e., at the limit between the so-called "quiet zone" and the oldest magnetic anomalies of the M sequence: they do not consider later meseta/stable Africa relative motion greater than 100 km; they regard the volcanoes of the Kelvin – Canary Island lineaments as the result of a residual deformation of both the African and North American plates. Dewey *et al.* (1973) adopted the same reconstruction at the end of the "quiet zone"; in their initial fit, however, they assume that the north of Africa was an assemblage of relatively independent blocks, otherwise they could not avoid a major gap North of the Kelvin – Canary Island lineament. In other words, the opening of the central Atlantic is accompanied north of the Canary Islands by stretching of the Middle- and High-Atlas domain, between 180 and 165 Ma; then, from 165–

150 Ma, the north African blocks moved away from North America faster than the African craton did, resulting in a dextral movement of the Moroccan meseta relative to the rest of Africa, a wider upper Jurassic "magnetic quiet zone" north of the Canary Islands than south of them, and compression of the Middle- and High-Atlas troughs. Then, north Africa has been sailing with stable Africa. The sedimentary assemblages and the intrusive–effusive volcanic material found there are, indeed, interpreted as evidence of restricted but true oceanic basins, closed in middle to upper Jurassic times.

Luyendyk and Bunce (1973) have reached a similar conclusion as to the initial block situation; however, even though part of the dextral motion of Morocco relative to Africa has taken place along the initial Triassic–Liassic distensive phases, most of it occurred in the Tertiary, according to the authors.

Le Pichon *et al.* (1977) favour a 250 km sinistral motion of Morocco relative to Africa after 150 Ma. Their model is based upon a reconstruction of the Atlantic at 150 Ma by fitting of the seaward edges of the magnetic quiet zones (MQZ): these are (south of the South Atlas fault) slightly different from the ones shown by Pitman and Talwani (1972) and, in addition, might not be strictly isochrons, as mentioned by the authors. The reconstruction shows that the African MQZ edge (north of the Canary Islands), rotated back toward North America, is too far west of the North American MQZ edge. Thus, the Moroccan block (with its associated MQZ) must have been located at that time to the east of its present position relative to stable Africa, by about 250 km, resulting, according to the authors, in a global sinistral motion of Morocco after 150 Ma, but prior to 80 Ma.

The divergent conclusions cited above call for a careful reexamination of the possible mobility of Morocco.

Geological Observations

Among the field geologists working in Morocco, there is general agreement on the limited compressive shortening of the Middle- and High-Atlas basins: 10–20 km (Choubert and Faure-Muret 1962; Mattauer *et al.* 1972; Proust 1973).

Mattauer *et al.* (1972) and Proust (1973) concluded that the main fault zone to the northeast of Agadir (Tizi n'Test fault) has experienced a significant (some 200 km) dextral wrench movement prior to the Triassic; later, the sinistral strike-slip motion has shown a 10 km magnitude. They clearly rule out a much larger amount. Petit (1976) reduced the slip to 50–100 km.

Michard *et al.* (1975) placed the same constraints on the motion along the Tizi n'Test fault, but did not exclude possible more complex motions of various continental small blocks, accounting for a global movement of a hundred of kilometres without closing a true oceanic Jurassic basin.

Atlantic History

In terms of kinematics, we face a three-plate problem: North America, Africa, and Iberia. Data are available for a reconstitution of the history of both pairs: North America – Africa and North America – Iberia. Whether Morocco acted as a fourth plate or not, has not yet been established unequivocally from available data: the great diversity of models, briefly outlined above, is an illustration. However, a new model for the Atlantic (Olivet 1978) places severe constraints on the possible mobility of Morocco.

The movement of Africa away from North America can be summarized in four episodes: firstly, a preliminary extensional episode affected the future continental margins and the Atlas domains, approximately from Triassic (230–220 Ma) to latest Liassic (180 Ma) (see Vogt (1973) and Van Houten (1977) for reviews). Secondly, Africa moved away from North America, then creating oceanic floor and resulting in a large global sinistral motion relative to Iberia (from 180 to 115–110 Ma, middle Cretaceous). Thirdly, from 115–110 to 90–85 Ma (earliest Coniacian), they were in their present relative longitudinal positions[2] and moved away from North America together. Fourthly, after anomaly 34, the last episode shows a slow relative convergence of Africa toward Iberia, the direction of which was either northwest or northeast.

The evolution (Fig. 2) just briefly outlined has been unravelled by restitution of the relative position of the plates by the fitting of oceanic isochrons at successive discrete times (Morgan 1968; Le Pichon 1968): these isochrons are most usually magnetic anomalies (Vine and Matthews 1963; Vine 1966). Of course, the fitting must be done separately for each pair of blocks. The following blocks have been considered: North America, Africa, Iberia. If, within these blocks, adjacent subplates have experienced relative movement, for a given time span, there has been a boundary between the two subplates at that time; fossil remnants of such a boundary are likely to be found in the ocean floor of corresponding age. Without taking into account the relative motion of the sub-

[2]More precisely, this means that the poles of relative motions of North America – Africa and North America – Iberia are close to each other on a great circle perpendicular to both flow lines.

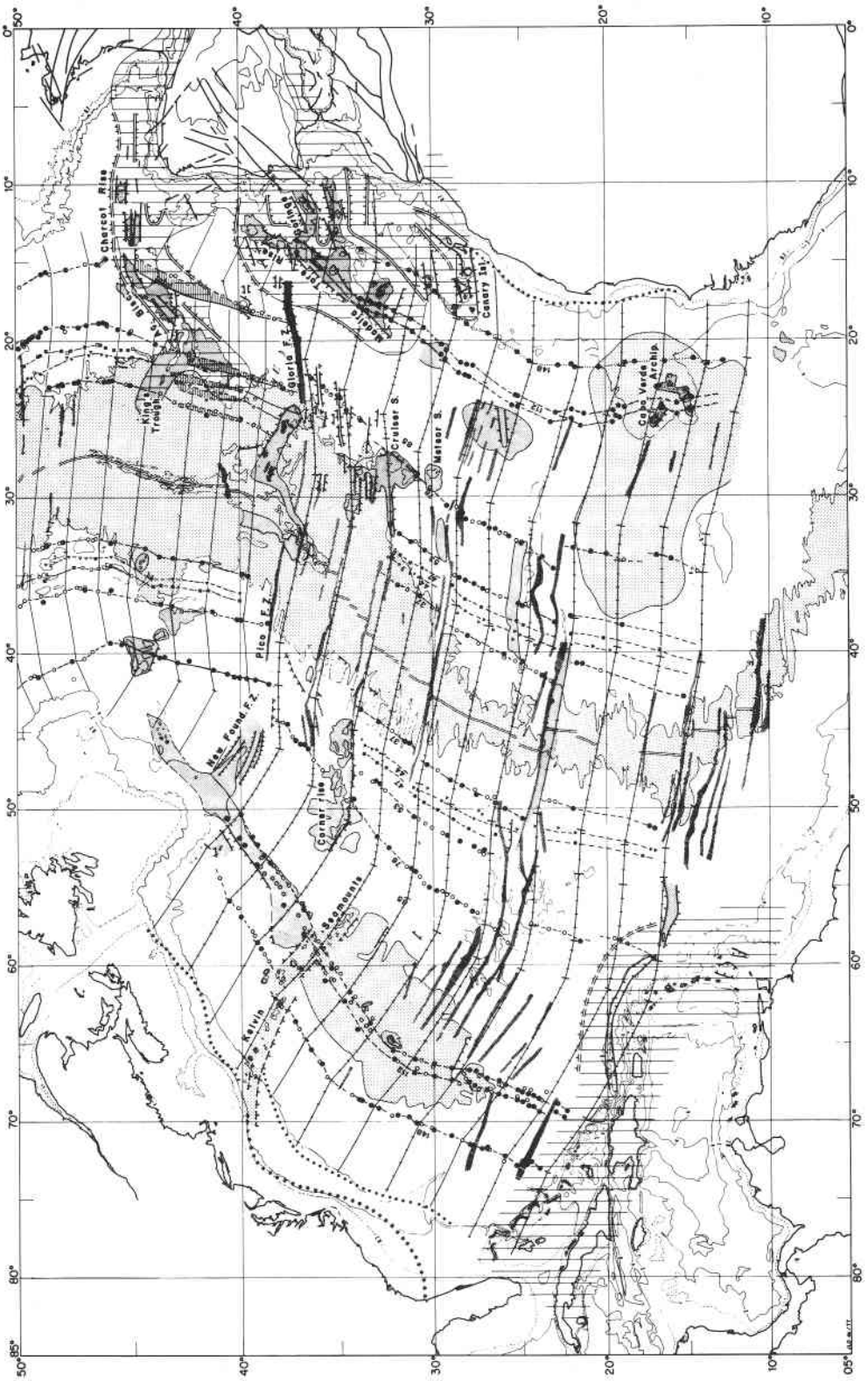


FIG. 2. Central Atlantic (after Olivet 1978). Solid symbols show the identified magnetic anomalies. Open symbols show the same anomalies after rotation. Figures refer to reconstituted isochron age in Ma (after Schlich 1975 and Schlich *et al.*, 1979). M22 = 148 Ma; J (M0–M3) = 112 Ma; 33 = 76 Ma; 24 = 53 Ma; 20 = 44 Ma; 13 = 37 Ma. Stars indicate the western E anomaly (Rabinowitz 1974) and the rotated eastern E anomaly (180 Ma). Flow lines are shown. Closely hatched areas: fracture zones; stippled areas: anomalous relief zones; loosely hatched areas: deformed oceanic zones in the vicinity of plate boundaries (Iberia–Europe; Africa-Iberia; South America; North America).

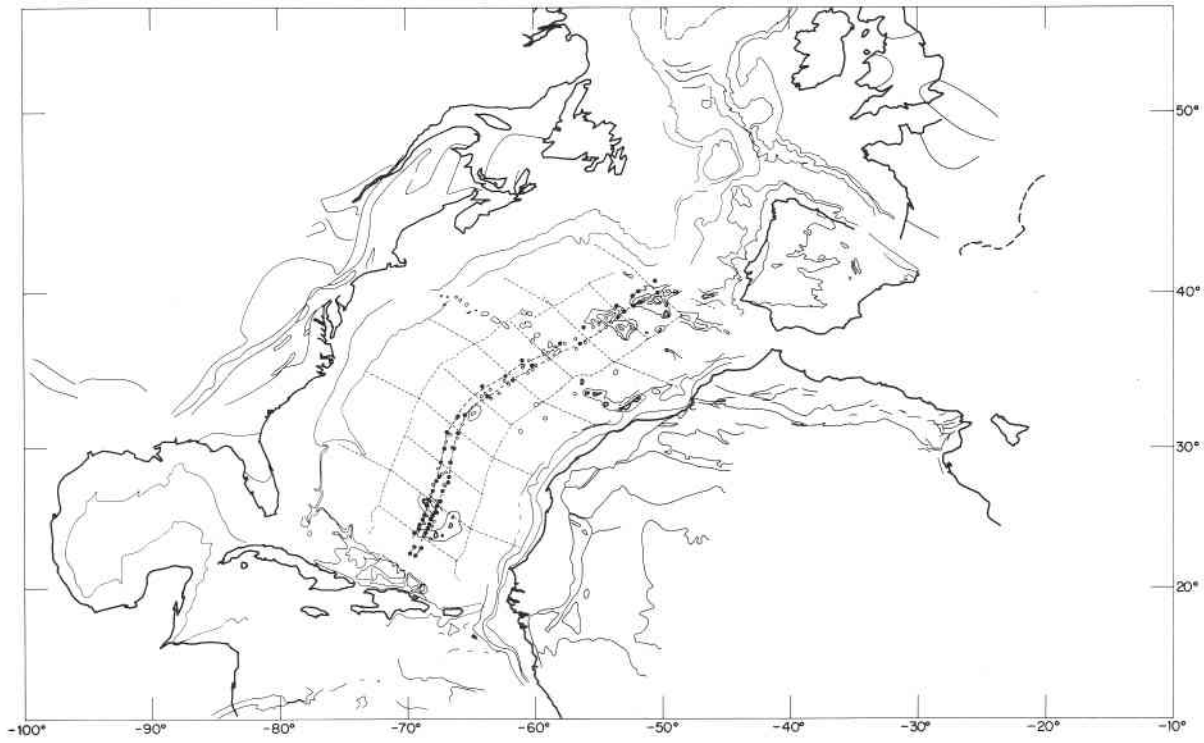


FIG. 3. Reconstruction at anomaly J time (ca. 112 Ma; Aptian), after Olivet (1978). Symbol meaning: see Fig. 2. Dashed lines are flow lines. Parameters of finite rotation: Eur./N.Am.: 89°8.5'N–103°8.5'W; angle 29.6°. Ib./N.Am.: 70°42.5'N–17°35.1'W; angle 49.9°. Af./N.Am.: 66°29.2'N–21°39.8'W; angle 55°.

plates, it will not be possible to fit isochrons older than the relative movement. An example is given by the Europe–Iberia pair of plates: in the present ocean, the once active boundary between the two plates is marked by King's trough, Azores-Biscaye-Charcot ridge, and the North Spanish trough. Another example is the still active boundary between Iberia and Morocco: Azores Islands, Gloria fault, Tore-Madeira rise, and Gorringe ridge complex.

Is there anything comparable in the present Atlantic for the pair Morocco–Africa? The South Atlas fault is, as already mentioned, aligned with an offset of the continental margin, itself prolonged by the volcanic axis of the Canary Islands. However, farther to the west, there is no apparent topographic anomaly between the Cruiser–Meteor volcano group and the Canary Islands.

Restitution at Anomaly 33

The anomaly 33 (Larson and Pitman 1972; Cande and Kristoffersen 1977), 75 Ma (Campanian), is the most easily recognizable isochron in the central Atlantic. Fitting the anomaly 33 isochrons (thus restituting the accreting axis at this time) suggests that no appreciable differential motion occurred between Morocco and Africa after 75 Ma.

Restitution at Anomaly J (Fig. 3)

Pitman and Talwani (1972) have recognized a prominent positive anomaly and coined it "J anomaly"; it turned out that this J anomaly was equivalent to anomalies M0 to M3 of the youngest part (ca. 110 Ma) of the Keathley sequence (Larson and Hilde 1975). In the Atlantic Ocean, it is also the age of anomalous topographic highs: buried ridge between Newfoundland fracture zone and Kelvin seamounts on the western side, Tore-Madeira rise and Cabo Verde archipelago on the eastern side. This is also the time at which ceases the left lateral global motion of Africa relative to Iberia and the start of approximately parallel movements of both the plates.

Restitution at Anomaly M22 (Fig. 4)

On both sides of the central Atlantic appears a zone of subdued magnetic anomalies, in contrast with the other oceanic areas; this zone is known as the "magnetic quiet zone" (MQZ) (Heirtzler and Hayes 1967; King *et al.* 1961); the anomalies show a peak-to-peak amplitude of 40–100 nT, whereas the same parameter in a "normal" oceanic environment ranges from 250–600 nT or even more. Seawards of the MQZ is found the Keathley magnetic sequence (Heirtzler and Hayes 1967; Vogt *et al.*

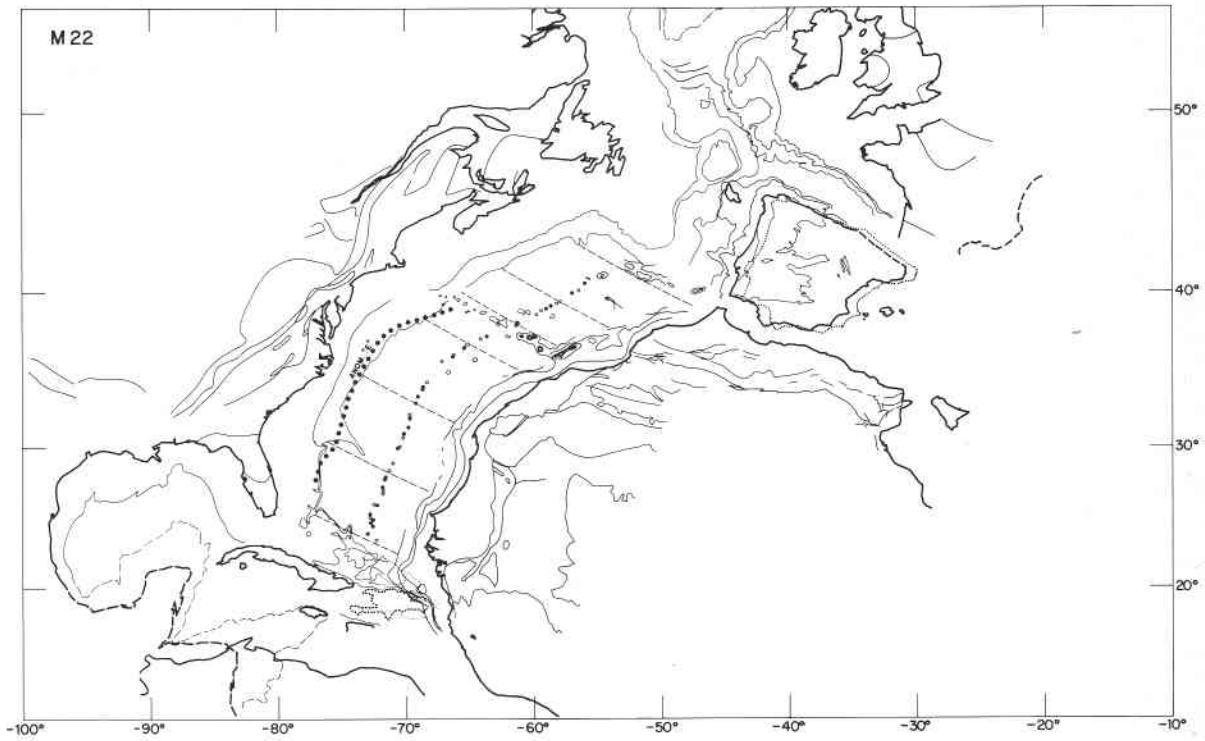


FIG. 4. Reconstruction at anomaly M22 time, after Olivet (1978). Symbol meaning: see Fig. 2. Parameters of finite rotation: Eur./N.Am. and Ib./N.Am. (cf. Fig. 7). Af./N.Am.: 65°34.8′N–18°20.1′W; angle 63.9°.

1969, 1970; Rona *et al.* 1970) which has been modeled, in terms of geomagnetic reversal chronology, by Larson and Pitman (1972). While several authors have regarded the limit between the MQZ and the Keathley sequence as an isochron and have proposed divergent conclusions (see, e.g., Pitman and Talwani 1972; Le Pichon *et al.* 1977), Larson and Hilde (1975) and Hayes and Rabinowitz (1975) have shown that the boundary spanned 5 Ma in time, from M22 to M25, and thus was not an isochron.

The latter authors have proposed an identification of anomaly M22 on the African side, which we have adopted. To determine the M22 anomaly on the western side, data available are: south of the Kelvin seamounts the abundant data (Vogt *et al.* 1970) have been reinterpreted following the procedure of Larson and Hilde (1975); north of the Kelvin seamounts, Barrett and Keen (1976) have come up with the most unique interpretation yet published, which turns out to be quite incompatible with the above identifications; we have thus reinterpreted the M22 anomaly (Figs. 5, 6). Then the accreting axis at the corresponding time (ca. 150 Ma) has been reconstructed. Once again, we do not see any possibility of having an appreciable motion along the Kelvin seamounts – Canary Islands boundary after anomaly M22 time.

In conclusion, the new reconstruction model implies that only limited movements of Morocco with respect to Africa after 150 Ma are possible; this is compatible with the observations of field geology and can account for the observed structures. However, a true independent meseta plate has not been in existence since Kimmeridgian times.

Possible Independent Motion of Morocco Prior to Late Kimmeridgian

The study of the kinematics of Iberia (Le Pichon *et al.* 1977; Olivet 1978) have shown the motion of intervening subplates. The opening of the Atlantic Ocean has proceeded from south to north, including the movement of Iberia: a right lateral motion of Morocco relative to Africa does not fit this simple model.

On the other hand, the apparent necessity of such a movement comes from the preopening reconstruction of the continental masses bordering the Atlantic Ocean. Most of the authors (see, e.g., Dewey *et al.* 1973; Luyendyk and Bunce 1973) assumed that the present-day continental margins were formed as an immediate consequence of the

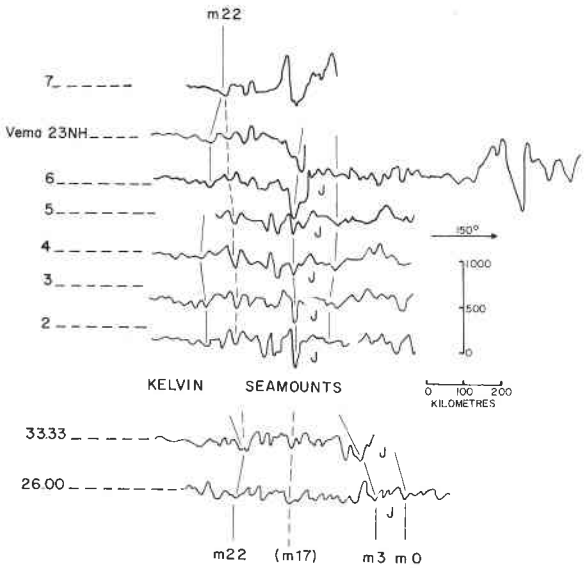


FIG. 5. Anomaly correlations; modified from Barrett and Keen (1976). Profiles 26.00 and 33.33 (south of Kelvin seamounts) are interpreted after Larson and Hilde (1975) and are used as reference. Profiles 2 to 7 (north of Kelvin seamounts; see Fig. 6 for location): the correlations of Barrett and Keen are shown in dashed lines; those of the present work are in solid lines. The latter interpretation of anomaly M3 corresponds to anomaly M17 of Barrett and Keen. Anomaly J (M0–M3) is the present work's identification.

Triassic–Liassic general distension. In their views, before this time the present-day continental margins must have been stuck against each other; with this philosophy in mind, the tightest fit they arrive at leaves a gap between Morocco and North America which can only be filled by decoupling the Moroccan meseta from stable Africa and moving it farther west against North America. This clearly implies a later right-lateral motion of Morocco relative to Africa.

Other authors (Le Pichon *et al.* 1977; Klitgord and Schouten 1977) have obtained a fit of Africa against North America such that the continental margins are contiguous even at the "latitude" of Morocco, the latter being attached to Africa in its present position. But with Africa positioned this way, the authors face difficulties in placing Iberia between Europe, Africa, and North America: Le Pichon *et al.* (1977) show that the only possible position leaves a considerable gap between the Grand Banks of Newfoundland and Iberia; they state that no reconstruction can be done, taking into consideration the known continental appendages of the continental masses (Galicia Bank, Flemish Cap, Ophan Knoll) without leaving such a gap. Sclater *et al.* (1977) work at a much larger scale and simply ignore the microcontinents (see also Grant 1975; Keen and Hyndman 1979; Kristoffersen and Talwani 1977).

An alternative view has been proposed by Olivet (1978) (Fig. 7). His preocean reconstruction, with plates kept rigid, shows a set of gaps, including the gap between Morocco and North America, but also between Iberia and North America and to the north of Iberia. The model relies upon the known extent of the evaporite basins off Iberia and Morocco (Montadert and Winnock 1971; Pautot *et al.* 1970). The age of the evaporites cannot be younger than Liassic as, in the adjacent continental basins, sediments are deposited in marine environments since the end of the Liassic (Montadert and Winnock 1971; Van Houten 1977). The basins' extent places constraints on the relative positions of continents at that time. Closing these basins' gaps implies much more than moving Morocco to the west: it needs actually a general disruption of southwestern Europe and North America. A possible wrench movement along the South Atlas fault must be included in this pre-Liassic context.

Mattauer (1973) and Arthaud and Matte (1977) have argued that all the large wrench faults of western Europe and western Africa have acted principally in "late Hercynian times," i.e., from upper Carboniferous to lower Permian. On the other hand, gaps in the reconstruction represent preserved primitive (with respect to the Atlantic Ocean) basins which are not isolated: north of the west Moroccan proto-basin, the un- or little-deformed Gulf of St. Lawrence basins are Permo-Carboniferous (Sherwin 1973); another example is the North Sea, of Permian age (Ziegler 1975).

We must then conclude that the kinematic evolution of the Atlantic Ocean does not support the argument in favor of Mesozoic (and Cenozoic) mobility of the Moroccan meseta relative to stable Africa.

Paleomagnetism

If it appears possible to calculate for "mobile Morocco," at a given time, a virtual geomagnetic pole (VGP) significantly different from that of "stable Africa," then the two blocks must have been independent at that time. On the other hand, if the two VGP's are not significantly different, there is no paleomagnetic reason to consider two blocks.

Data relative to the Liassic are available for both "stable Africa" (see, e.g., Briden *et al.* 1971; Hailwood and Mitchell 1971; Dalrymple *et al.* 1975; Bonifay *et al.*, manuscript in preparation; Sichler *et al.*, manuscript in preparation) and "mobile Morocco" (Michard *et al.* 1975; Westphal 1976). Until recently data on stable Africa were scarce, so

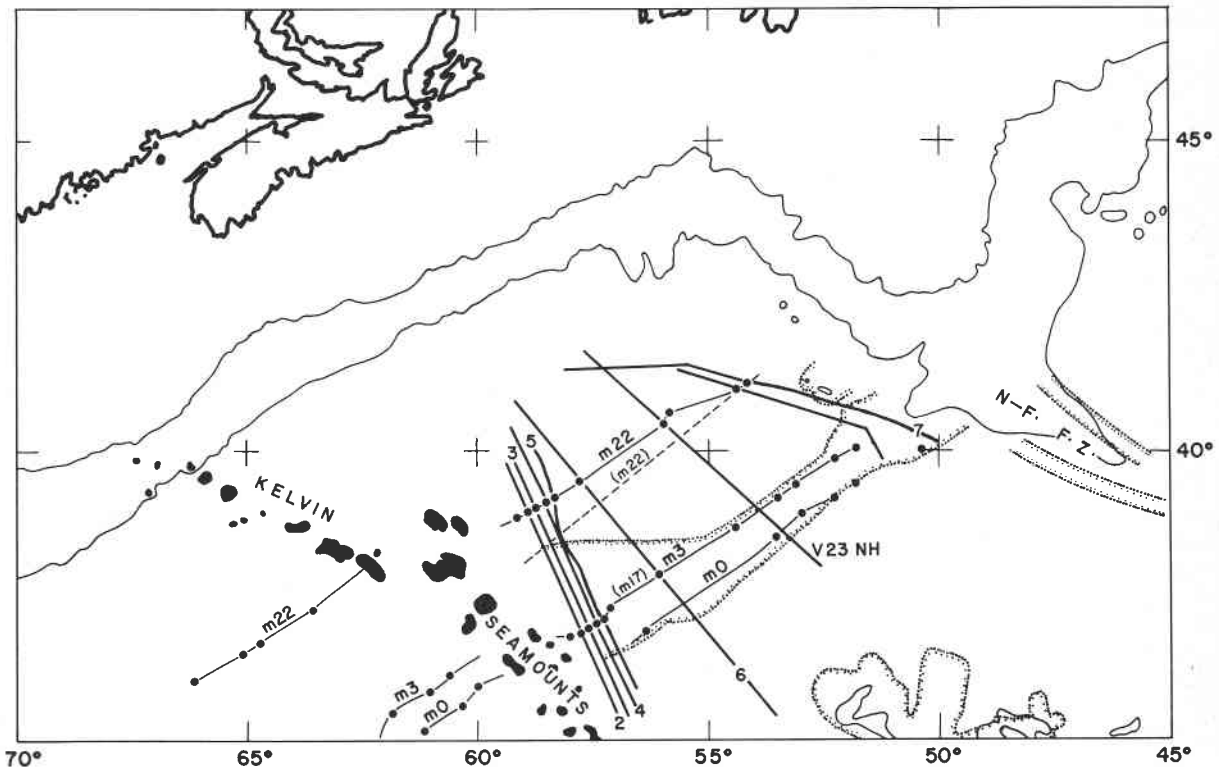


FIG. 6. Locations of profiles of Fig. 5, modified from Barrett and Keen (1976). Identifications of latter authors are shown within parentheses. Complementary identifications have been done using various sources. Stippled areas indicate zones of anomalously high basement, in particular, the ridge at anomaly J location; after Oivet (1978).

that to determine the correspondant VGP, data from South Africa had to be used. Owing to the angular distance between Morocco and South Africa, the comparison between the two VGP's suffered limitation; dispersion of VGP from data relative to entire stable Africa was eventually large (and impossible to reduce), owing to the influence of the possible nondipole component of the geomagnetic field, which is simply ignored in the procedure. Recent data have permitted the restraint of the "stable Africa" reference region and the inclusion of samples from zones close to mobile Morocco. At the same time, the influence of any contingent deformation between West Africa and South Africa, difficult to evaluate, is excluded (Fig. 8).

Table 1 summarizes the results of paleomagnetic studies on five large regions within the African craton. The regions' mean poles are obtained from the sites' poles without using the mean directions of each region. Some of them have then been recomputed and are eventually different from what has been published. The pole for "stable" Morocco is somewhat peculiar: as the 21 sites of Hailwood and Mitchell (1971) yield directions not significantly different (at the 95% confidence level) from the results on the same region obtained by Bonifay *et al.* (manuscript in preparation) on 21 sites, only one population of 42 sites has been considered and the computations conducted consequently.

Allocating a unit weight to each sampled region (Fig. 9), the computation yields a VGP at 71.1°N and 238.9°E with an 95% confidence cone angle of 6.1° for stable Africa. The same calculation conducted with a unit weight affected to each site yields a confidence angle much lower (2° with 103 sites); we feel such a procedure is questionable: considering homogeneous a site population dispersed in five regions from Liberia to Morocco appears hazardous. Ages of samples from south Morocco, Hodh, and Liberia have been determined by the K–Ar method: 184 ± 2, 187 ± 5, and 182 ± 10 Ma respectively. Even though the uncertainty intervals overlap quite satisfactorily, they are long enough for the geomagnetic field to experience sensible variations from one magmatic episode to another.

Giving a unit weight to a site makes the general mean VGP shift toward the VGP of the best sampled region; then the more precise result is chiefly

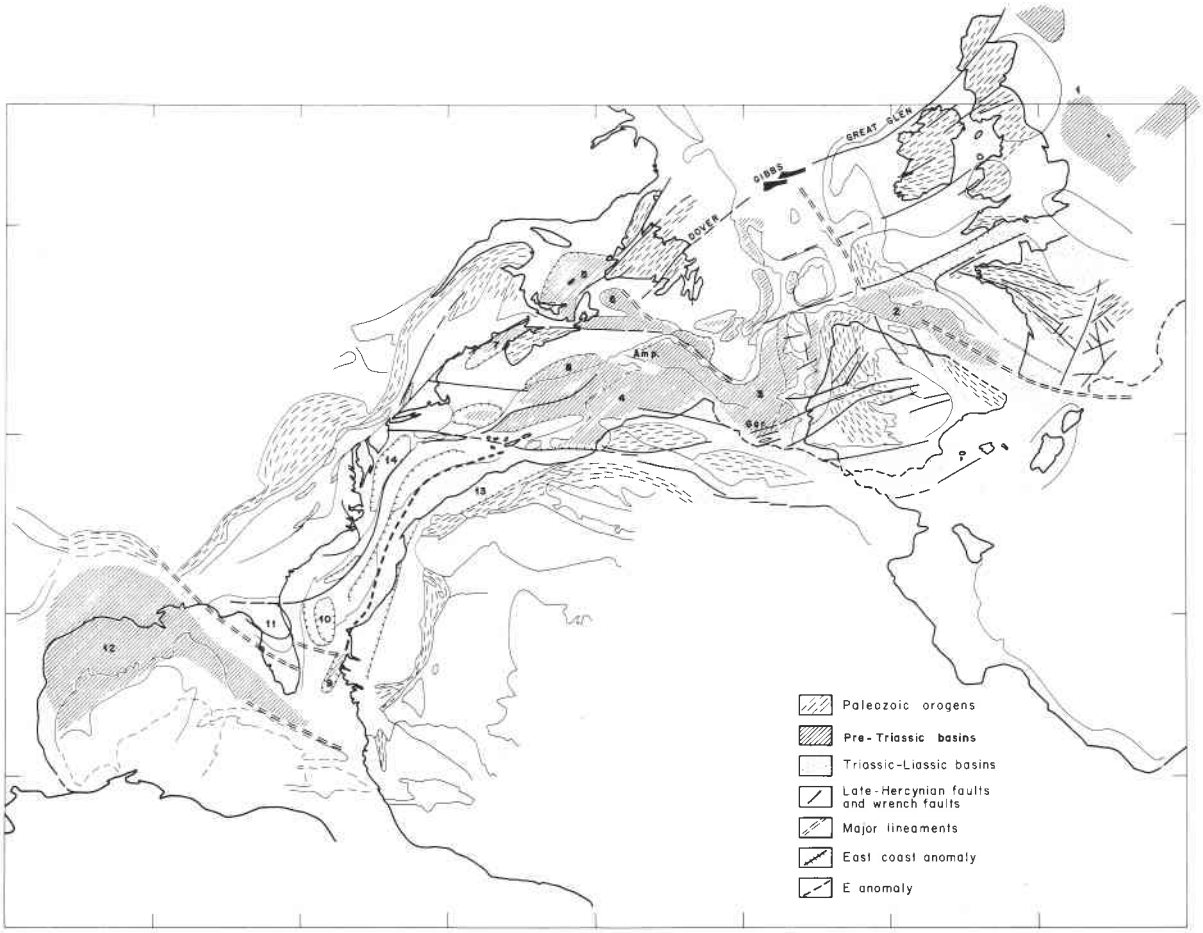


FIG. 7. Reconstruction at Liassic times, after Olivet (1978). (1) North Sea; (2) proto-gulf of Biscaye; (3) west Galicia, Tagus, east Newfoundland; (4) Seine basin (dashed line: limit of evaporites; Amp.: Ampère-Bank; Gor.: Gorringe Bank); (5) Magdalena; (6) Sydney; (7) Acadia; (8) Georges Bank; (9) South Senegal; (10) Blake plateau; (11) Survanee; (12) Gulf of Mexico; (13) Rio de Oro; (14) Baltimore Canyon. Parameters of finite rotation: Eur./N.Am.: 89°8.5′N–103°8.5′W; angle 29.6°. Ib./N.Am.: 73°54.2′N–16°36.1′W; angle 49.1°. Af./N.Am.: 66°15.3′N–17°6.7′W; angle 74.1°.

representative of that region, eventually affected by the local paleoanomaly of the geomagnetic field.

The mean pole of the five West African regions (unit weight to each region) (Table 1) is determined with medium precision (6°). Two regions yield contributions which deviate from the mean pole: (*i*) the Freetown complex ($\delta_1 = 10.5$): a small number of samples have been taken and the age determinations are uncertain, possibly because of argon pollution by the embedding rocks (Dalrymple *et al.* 1975) (see Table 1); (*ii*) stable Morocco ($\delta_1 = 6.2$): the above remarks do not apply to this region. As a significant tectonic motion between the different West African regions is excluded a priori, we feel that the discrepancy is due to a slight time lag in the magmatic episodes and (or) regional paleoanomalies.

The angular difference between the VGP's of "mobile" Morocco and the West African craton is 6.1° (Table 1): we are just at the limit of what is statistically significant. The difference could be accounted for by the reasons invoked above for "stable Morocco." The VGP of the former has been computed with a unit weight to the sites. It could have been done with a unit weight to each of the four regions sampled; there is indeed a significant (at the 95% confidence level) difference between the results from the meseta and from the western and central Atlas (Table 2). With the second procedure, the angular distance between the VGP's of "mobile Morocco" and stable Africa is 4.5°, well within the uncertainty (11°) (Table 2), even though, in this case, the precision estimation is probably pessimistic.

The angular distance between the two VGP's could be reduced further by including in the com-

TABLE 1. Results of paleomagnetic studies

Region	Number (Fig. 8)	Number of sites	K	A_{95}	VGP Latitude (°N)	VGP Longitude (°E)	δ_1	δ_2	References
"Stable" Morocco Anti-Atlas	1	42	76.5	2.5°	65.1	243.6	6.2°	6.9°	Hailwood and Mitchell (1971); Bonifay *et al.* (in preparation)
North Mauritania Hank	2	13	99.0	4.1°	69.4	232.0	2.9°	2.3°	Sichler *et al.* (in preparation)
South Mauritania Hodh	3	13	46.5	6.1°	71.4	240.2	0.5°	1.4°	Sichler *et al.* (in preparation)
Sierra Leone Freetown	4	10	70.0	5.7°	81.5	231.3	10.5°	10.1°	Briden *et al.* (1971)
Liberia	5	25	31.9	5.2°	68.2	242.2	3.1°	3.9°	Dalrymple *et al.* (1975)
"Mobile" Morocco	6	34	17.3	6.0°	71.9	219.5	6.1°	5.1°	Michard *et al.* (1975); Westphal (1976)
Mean (first five regions)	I	5	156.4	6.1°	71.1	238.9	s 6.5°		
Mean (six regions)	II	6	164.4	5.2°	71.4	235.8	6.3°		

NOTES: K = precision parameter; A_{95} = cone angle at 95% confidence; δ_1 = angular distance between the region's VGP and the mean VGP I; δ_2 = angular distance between the region's VGP and the mean VGP II; $s = [\Sigma \delta_i^2/(N-1)]^{1/2}$ angular standard deviation for the two means. Mean direction comparison (Watson 1956). To compare mean directions obtained with diverse populations, it is necessary that these populations show dispersion statistically not different; this can be tested by using the Fisher–Snedecor tables. When the dispersions are comparable, the same tables can be used for testing the identity of the mean directions. When the dispersions are different, one can only compare the mean direction with a free-of-error direction, using the 95% confidence cone within which the mean direction is defined (the "free-of-error" direction can be another mean direction).

TABLE 2. Virtual geomagnetic poles for "mobile" Morocco

Reference	Region	No. of sites	VGP Latitude(N)	VGP Longitude (E)	A_{95}
(Westphal 1976)	Meseta	8	68°	201°	10°
	Atlas, north	18	74°	215°	9°
	Atlas, centre and west	4	62°	240°	26°
	Atlas, south	4	76°	247°	23°
Mean of four regions			71°	225°	11°

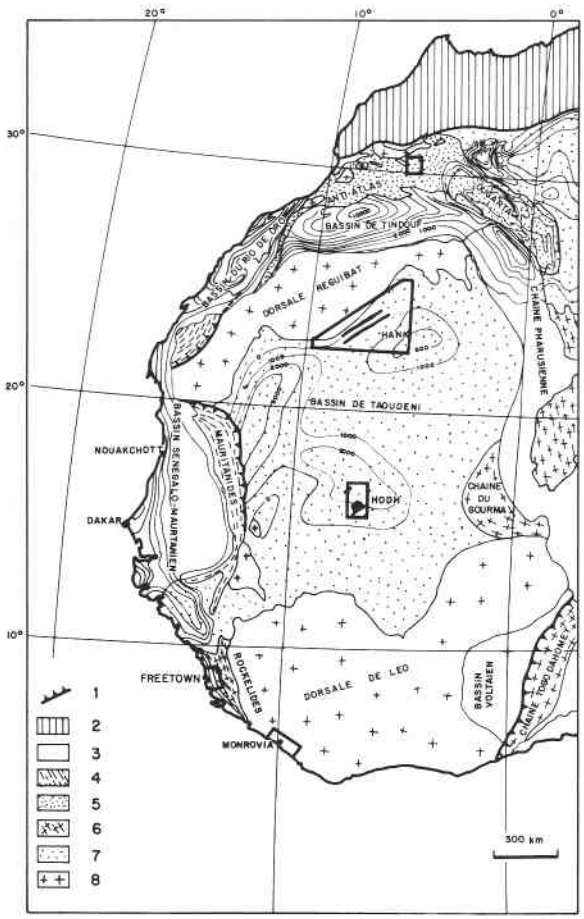


FIG. 8. Simplified geological map and paleomagnetic sample sites. (1) Overthrust fault. (2) Alpine influenced domain. (3) Post-Paleozoic basins. (4) Hercynian chain. (5) Upper Precambrian and Paleozoic cover, folded in Hercynian times. (6) Panafrican domain (650–550 Ma). (7) Upper Precambrian and Paleozoic basins. (8) Precambrian basement (1000 Ma). Boxes indicate the five regions of the west African craton which have yielded paleomagnetic data used in the present study.

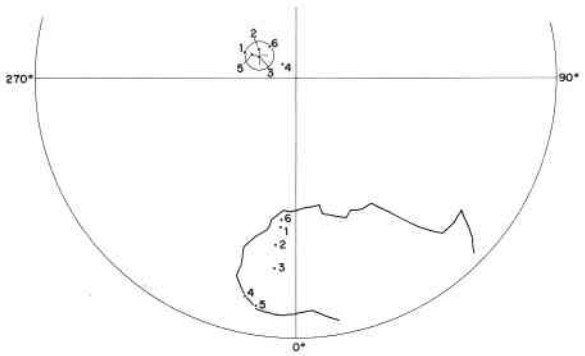


FIG. 9. Liassic virtual geomagnetic poles for the five west African regions (see Fig. 8) with the 95% confidence circle. Pole number 6 is for "mobile" Morocco. Stereographic projection.

putation of the Moroccan VGP the data from Hailwood and Mitchell (1971) (six sites; VGP at 53°N, 261.5°E, A_{95} = 24°). However, the dispersion of their age determinations is large and the authors do not give any stratigraphic piece of evidence, whereas Westphal (1976) has put forward a very useful fine stratigraphy.

Finally, paleomagnetic data fail to prove that Morocco has been independent of Africa in Liassic time. However, the resolving power of the paleomagnetic method is here only 5–6°: so that it is not possible to appreciate relative rotations of that magnitude or lower; thus limited Liassic motions advocated by field geologists along major faults within the South Atlas zone (see, e.g., Proust *et al.* 1977) are not ruled out: the small curvature of the zone and the limited amount of global slip invoked yields a very small angular amplitude.

It is noticeable that including all data ("mobile Morocco" plus stable Africa) in the calculations leads to a mean VGP with better precision parameters (Table 1).

Conclusion

It has been shown above that, at least from Jurassic times, neither reconstruction geometry nor slight divergence between the Liassic paleopoles require a decoupling between Africa and Morocco. Other authors have suggested that such a conclusion holds from the Autunian on for some of them (Daly and Pozzi 1976), from the Carboniferous on

for others (with a difficulty during lower Permian) (Martin *et al.* 1978).

It is not our intention to claim that Morocco and Africa have always behaved as a unique block. In the field, evidence of strike-slip movements are clearly present. However, the present study rules out any movement along the South Atlas fault, from the Jurassic, which could compare with major boundaries, such as, for example, the Azores–Gibraltar lineament (between Africa and Iberia) or the North Pyrenean fault (between Iberia and Europe).

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